

AKASH MAURYA

AI/ML Engineer Intern | Computer Vision Engineer Intern

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SUMMARY

AI/ML undergraduate with hands-on research and internship experience in deep learning, computer vision, and image super-resolution. Skilled in PyTorch, Python, OpenCV, and Flask, with proven ability to implement, train, and evaluate CNN- and Transformer-based models (EDSR, ESRGAN, Real-ESRGAN, SwinIR, HAT, DRCT) across multiple public and custom datasets. Research exposure to self-supervised learning (SimCLR), biometric image identification, and real-time embedded systems. Seeking an AI/ML internship to apply strong model development, data engineering, and evaluation skills to production-grade AI solutions.

SKILLS

Languages & Core: Python, C/C++, SQL, Data Structures, Algorithms, OOP

ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas, Matplotlib

Techniques: Deep Learning, Computer Vision, Image Super-Resolution, Self-Supervised Learning (SimCLR), CNN, RNN, LSTM, GRU, Transformers, Regression, Classification, Clustering (K-Means), Model Evaluation, Feature Engineering, Data Preprocessing

Models: EDSR, ESRGAN, Real-ESRGAN, SRCNN, FSRCNN, VDSR, SwinIR, HAT, DRCT, SRTTA

Tools & Platforms: Flask, Git, GitHub, Linux, Jupyter Notebook, VS Code, Arduino, ESP32

EDUCATION

B.Tech in Artificial Intelligence - Bennett University

Aug 2024 - Present

CGPA: 8.71/10 | Expected Graduation: May 2028

Class XII (PCM) - UP Board

Apr 2022 - Mar 2023

Result: 88%

Class X - UP Board

Apr 2020 - Mar 2021

Result: 92.17%

EXPERIENCE

Summer Research Intern (Remote, Part-Time) - ViSecure Systems Pvt. Ltd.

Jun 2026 - Aug 2026

- Trained and benchmarked 10+ image super-resolution architectures - EDSR, VDSR, ESRGAN, Real-ESRGAN, SRTTA, DRCT, SwinIR, and HAT - on multiple datasets including Vehicle-10, PlantVillage Disease, and a custom combined HR/LR dataset.
- Engineered unsupervised super-resolution pipelines by applying self-supervised representation learning with SimCLR and contrastive loss alongside the EDSR, ESRGAN, and HAT architectures.
- Evaluated model performance using quantitative metrics (PSNR, SSIM) and qualitative visual analysis across benchmark datasets including RealSR, DIV2K, and Urban100.
- Implemented, configured, and executed research repositories for multiple super-resolution models across different datasets using Git for version control.
- Analyzed model architectures, training and inference pipelines, and loss functions to understand their impact on reconstruction quality and super-resolution performance.
- Tech: Python, PyTorch, OpenCV, Deep Learning, Computer Vision, Git, Linux, Jupyter Notebook.

Research Intern (Remote, Part-Time) - Ashoka University

Jun 2026 - Sep 2026

- Conducted research on animal biometric identification under the Simons Fellowship Program, supervised by Prof. Anurag Agrawal.
- Captured and analyzed buffalo iris images under varied lighting conditions, including early-morning sessions, to reduce specular reflection and improve image quality for biometric feature extraction.
- Identified corneal specular reflection as a key limitation in buffalo iris imaging and evaluated capture-angle and lighting adjustments to mitigate it; determined that the lack of a large, labeled buffalo iris dataset was the primary blocker to model development and sourced an available public dataset to continue the analysis.
- Extended the research direction toward snake biometrics, adapting image-based biometric identification methods for individual snake identification.
- Conducted image analysis, dataset preparation, literature review, and experimental investigation as part of the ongoing research.

General Secretary & Co-Founder - RoboGenesis Club, Bennett University

Aug 2025 - Present

- Co-founded and lead a robotics and AI club with 60+ student members, managing a core team of 23 and organizing technical workshops, hackathons, and student onboarding sessions.
- Organized and ran a 24-hour industry-driven hackathon for the Club and School of AI, drawing 60 teams and 140+ participants from 14+ institutions.
- Mentor teams in developing end-to-end projects using Python, OpenCV, Arduino, and IoT systems.
- Drive student participation through structured mentorship, hands-on sessions, and technical activities.

PROJECTS

Thermal Image Super-Resolution Research - Team

Jan 2026 - May 2026

- Evaluated deep learning-based super-resolution models including SRCNN, FSRCNN, VDSR, and EDSR for thermal image enhancement.
- Analyzed SwinIR and transformer-based super-resolution models, examining architectures, inference pipelines, and reconstruction performance on low-resolution infrared images.
- Built practical expertise in model training, inference, dataset preprocessing, image reconstruction, and evaluation using PSNR and SSIM metrics.
- Tech: Python, PyTorch, OpenCV, TensorFlow.

Autonomous Line-Following RC Car - Team

Sep 2025 - Mar 2026

- Developed a line-following robot using IR sensors and PID control with obstacle detection and basic sensor fusion.
- Tech: Arduino, ESP32, C/C++.

ImageLab - K-Means Image Compressor - Team

Sep 2025

- Built an image compression system using K-Means clustering, reducing image size by 40-60% while maintaining perceptual quality.
- Tech: Python, OpenCV, scikit-learn.

KrishiShram - Farmer Labour Management Web App - Independent

Jan 2025

- Developed a full-stack web application to digitize labour records and automate wage calculation, reducing manual effort by 60% through automation.
- Built RESTful APIs using Flask and SQLite for efficient data handling.
- Tech: HTML, CSS, JavaScript, Python, Flask, SQLite, Git.

CERTIFICATIONS

- Google AI Essentials - Coursera (Google) - Jul 2025
- Prompt Engineering - Infosys Springboard - Sep 2025
- Neural Networks and Deep Learning - Coursera (DeepLearning.AI) - Feb 2026

AWARDS & ACHIEVEMENTS

- INSPIRE Scholarship (DST) for academic excellence
- Academic Silver Medal for top 10 position in Class XII (UP Board)